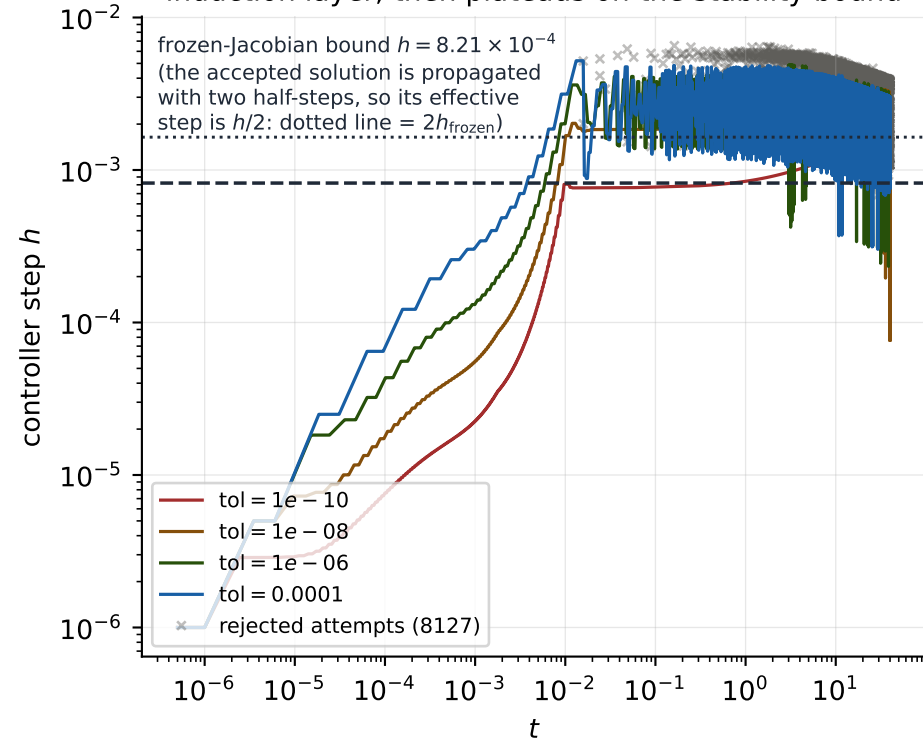
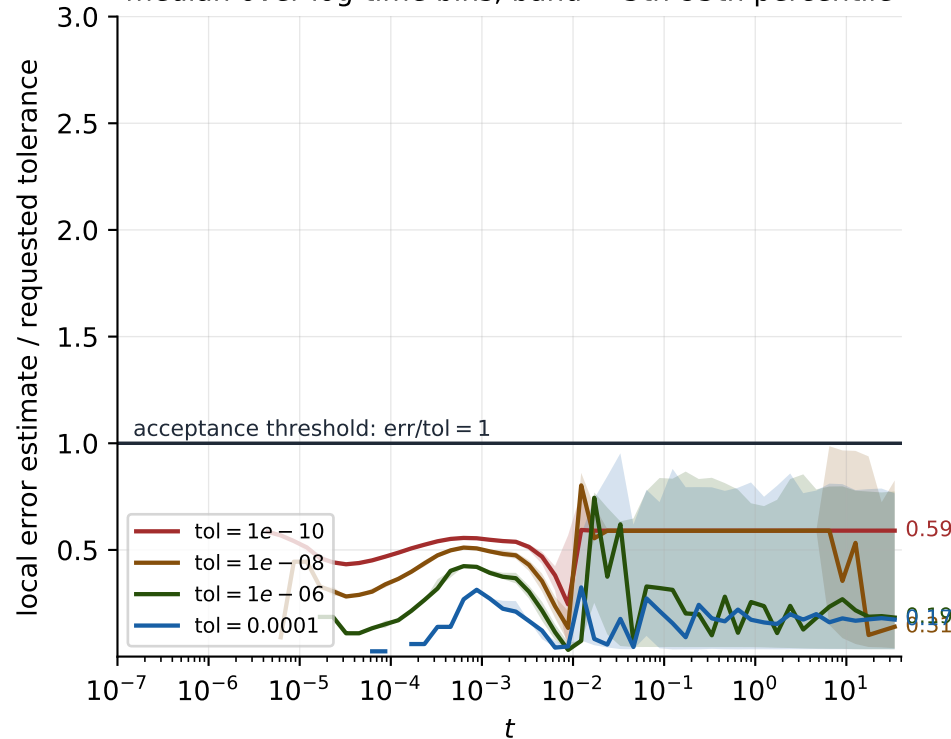


F11 RK4 step-doubling controller: step size, error tracking, and a norm that fails

(a) The step size is not constant: it collapses through the induction layer, then plateaus on the stability bound



(b) The controlled quantity does track the tolerance median over log-time bins, band = 5th-95th percentile



(c) What the tolerance buys globally - and the norm that silently loses the stiff mode

